

Tushar Supe

Senior Backend and AI/ML Engineer

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PROFESSIONAL SUMMARY

Backend and AI/ML engineer with 3 years of experience building and operating production computer vision and applied AI systems for an AI-powered image platform serving automotive marketplaces across the United States, Canada, and India. Progressed from sole AI/ML engineer at an early-stage startup to technical lead, owning the full applied-AI lifecycle while managing a team of AI and backend engineers. Cut third-party inference costs by 75%+ and eliminated them entirely with in-house models, and reduced core API latency by over 99%. Holds a Bachelor's degree in Programming and Data Science from IIT Madras and a Bachelor's degree in Mechanical Engineering from VNIT Nagpur.

WORK EXPERIENCE

Senior Backend and AI/ML Engineer

July 2023 – Present

HelloRamp – Bengaluru, India

AI-powered car and bus imaging platform

Serves automotive marketplaces, bus operators, and 100+ U.S. dealership groups. Joined a six-person startup as a backend engineer and became its sole AI/ML engineer, independently owning the computer vision roadmap, backend platform, and cloud infrastructure end-to-end before progressing into a technical lead role.

Computer Vision and Applied AI

- Built an end-to-end virtual-studio car imaging pipeline: background removal, orientation and tilt correction, AI-generated shadow/reflection, license plate detection and anonymization, and studio-background placement, fusing 15+ production CV models to output photorealistic images at 97–99.5% accuracy or IoU.
- Scaled this pipeline into a high-throughput production system processing roughly a quarter-million images per month with 8x burst capacity, while cutting per-image processing time by 70% and pipeline memory footprint by over 95% through threading, caching, and alignment-logic optimization.
- Extended the pipeline to bus imagery and shipped a thin-part detection model, raising detection accuracy from 60% to 85% while cutting inference latency by roughly 4x and container size by roughly 15x through model and runtime optimization (ONNX export, image slicing, custom post-processing).
- Reduced model training compute cost by 75% through GPU tier optimization and custom training pipeline improvements, and cut distributed-GPU training wall-clock time from roughly 3 months to under a week through parallelization and pipeline redesign.
- Automated 3D synthetic-data generation (Blender, distributed multi-GPU rendering), cutting a 450+ day rendering workload down to roughly a week and enabling continuous scaling of training data.
- Engineered an interactive 360-degree car spin generation pipeline from walkaround imagery, evaluating and applying multiple novel-view-synthesis and multi-view-geometry approaches spanning classical structure-from-motion, neural rendering, and video stabilization.

Backend, Cloud, and Platform Engineering

- Led a two-phase authentication performance overhaul on a core production API, cutting response time by over 99% to consistently under 100ms through database indexing, caching, and query redesign.
- Established MongoDB schemas, aggregation pipelines, and indexing strategy for a document database layer supporting multiple microservices, and built the asynchronous task queue infrastructure behind it, resolving recurring worker crashes, memory exhaustion, and message-duplication issues under production load.
- Drove a company-wide cost-reduction program – in-house model hosting, database and API cost optimization, automated stale-data cleanup, and inference/pipeline efficiency work – delivering over \$64,000 per year in combined recurring savings, plus a separate one-time \$2,000+ saving from GPU training optimization.
- Cut GPU inference hosting cost by roughly 7x by architecting a self-hosted, multi-model GPU inference platform across two cross-cloud migrations, reducing reliance on paid managed ML hosting.
- Implemented a GCP AI inference gateway across 30+ in-house inference endpoints with a Redis-based rate limiter and custom metrics-driven autoscaling, handling roughly 1,000 requests per 20-second window while optimizing for cost-efficiency over raw throughput to match real production load and minimize idle spend.
- Owned end-to-end production incident response and database reliability at scale – including a multi-month retention cleanup that reduced one major collection's size by roughly a third – across application, database, task queue, and third-party API layers as the platform's primary infrastructure engineer.

Applied R&D

- Evaluated and prototyped generative-model approaches for background inpainting, reflection removal, and image harmonization, training custom models for each.

Team Leadership and AI Product Delivery

- Advanced from sole AI/ML engineer to technical lead at a sub-10-engineer startup: hired and mentored four backend and AI engineering interns, converted two to full-time AI engineers, and continued leading those engineers alongside three additional interns – while still independently shipping individual technical work end-to-end.
- Own the AI and computer vision technical roadmap and technical hiring for AI engineering roles; review and merge pull requests across multiple repositories; serve as primary technical point of contact for major enterprise accounts.
- Currently steering a team building an automated image quality-check and auto-correction system that flags issues via AI and routes unresolved cases to human review.
- Guided a team through delivery of an automated social media content pipeline (AI avatar-video, script, and voice generation with multi-platform posting and comment-reply agents) and shipped internal analytics dashboards covering image quality, volume, failure rate, and team performance.

Maintenance Engineer, Graduate Engineer Trainee

2019 – 2020

Reliance Industries Limited

Mechanical plant maintenance at a large-scale manufacturing facility.

EDUCATION

Bachelor of Science, Programming and Data Science

2020 – 2023

Indian Institute of Technology (IIT), Madras. CGPA: 9.2/10

Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems

Bachelor of Technology, Mechanical Engineering

2015 – 2019

Visvesvaraya National Institute of Technology (VNIT), Nagpur. CGPA: 8.12/10

PERSONAL PROJECTS

Automated Job Application Agent – Personal Project

Agentic system automating the end-to-end job search workflow: parses job descriptions, matches them against a structured candidate profile, generates tailored application materials, and completes submission steps through browser automation. Built with LangGraph/LangChain for agent orchestration, retrieval-augmented generation over candidate and job data, tool-calling and MCP integration, and a FastAPI/PostgreSQL/pgvector/Redis backend orchestrated with n8n and Playwright.

SKILLS AND TOOLS

Languages: Python, Bash/Shell, JavaScript

Machine Learning and Computer Vision: Deep learning model development and training, distributed multi-GPU training (PyTorch, AMP/FP16), computer vision (segmentation, detection, classification, matting), object detection (YOLO-family), generative modeling (GANs, diffusion models, CNNs), inference acceleration (ONNX), synthetic data generation, OpenCV, scikit-learn, 3D reconstruction and novel view synthesis (COLMAP, GLOMAP, Gaussian Splatting, NeRF/NeRFStudio), multi-view geometry and feature matching (SIFT, SuperPoint, SuperGlue)

AI/LLM and Agentic Systems: LLM APIs (Claude, OpenAI, Gemini), LangGraph, LangChain, Retrieval-Augmented Generation (RAG, Agentic RAG), multi-agent systems, tool calling, Model Context Protocol (MCP), AI avatar/voice/video generation pipelines

Cloud and Infrastructure: AWS (EC2, S3, Lambda, SQS, CloudFront, SageMaker, Secrets Manager), GCP (Compute Engine, Cloud Run, Cloud Build, Cloud Tasks, Vertex AI), self-hosted GPU inference platforms, Docker, GitHub Actions, Nginx, Blender (bpy) for synthetic 3D data

Databases and Backend: MongoDB, MongoDB Atlas, PostgreSQL, pgvector, Redis, FastAPI, schema design and indexing, REST API development, asynchronous task queues (Celery), backend service architecture

Automation and Tooling: n8n, Playwright, webhooks, LangSmith (observability)

Practices: Git version control, code review, technical hiring and team leadership, mentorship, cost optimization and infrastructure governance, production incident response, CI/CD